

Mohan Krishna G

Machine Learning Engineer | Generative AI & LLM Engineering

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 6+ years of experience building, deploying, and supporting production machine learning and Generative AI solutions across enterprise, semiconductor manufacturing, product analytics, and supply-chain environments. Strong expertise in Python, SQL, PyTorch, Scikit-learn, XGBoost, deep learning, NLP, LLMs, Retrieval-Augmented Generation (RAG), embeddings, vector search, and predictive modeling. Experienced across the ML lifecycle, including data preparation, feature engineering, experimentation, model evaluation, deployment, monitoring, and CI/CD using AWS, SageMaker, MLflow, Docker, Kubernetes, Spark, Databricks, and Kafka. Proven ability to collaborate with Data Science, MLOps, Data Engineering, Product, and Software Engineering teams to move AI solutions from experimentation into reliable production environments.

TECHNICAL SKILLS

- Programming & Software Engineering:** Python, SQL, Pandas, NumPy, REST APIs, FastAPI, Git, Unit Testing, CI/CD, Microservices
- Machine Learning & Deep Learning:** PyTorch, Scikit-learn, XGBoost, Classification, Regression, Neural Networks, Deep Learning, Time-Series Forecasting, Anomaly Detection, Feature Engineering, Model Evaluation, Error Analysis, Statistical Modeling
- Generative AI & NLP:** LLMs, Retrieval-Augmented Generation (RAG), Hugging Face Transformers, Embeddings, Vector Search, Semantic Search, Prompt Engineering, NLP, LangChain, LangGraph, AI Agents, Agent Orchestration
- LLM Evaluation & Retrieval:** RAG Evaluation, Retrieval Evaluation, Similarity Search, Metadata Filtering, Groundedness, Faithfulness, Answer Relevance, Hallucination Detection
- Vector Technologies:** FAISS, Pinecone, pgvector, Vector Databases, Embedding Search, Semantic Retrieval
- MLOps & LLMops:** MLflow, Experiment Tracking, Model Registry, Model Deployment, Model Serving, Batch Inference, Real-Time Inference, Model Monitoring, Data Drift, Model Drift, Automated Validation, AI Observability, Docker, Kubernetes, CI/CD
- Cloud & ML Platforms:** AWS, SageMaker, S3, Lambda, Glue, EMR, Redshift, Bedrock, Databricks, Azure
- Big Data & Streaming:** Spark, PySpark, Kafka, Databricks, Delta Lake, Airflow, Batch & Stream Processing
- Analytics:** Product Analytics, KPI Development, Root-Cause Analysis, Power BI, Tableau

WORK EXPERIENCE

AI/ML Engineer | Adobe May 2025 – Present | USA

- Architected Python and PySpark machine learning workflows on AWS, transforming high-volume product and customer data into reliable datasets for experimentation, predictive modeling, and enterprise AI applications.
- Built Generative AI solutions using LLMs, Retrieval-Augmented Generation (RAG), embeddings, semantic search, and vector retrieval to improve contextual knowledge discovery and response quality across intelligent product experiences.
- Engineered reusable feature-generation pipelines for structured, unstructured, and behavioral data, standardizing model inputs across development, evaluation, inference, and downstream machine learning workflows for consistent processing.
- Evaluated RAG applications through retrieval quality, response relevance, grounding, and model behavior analysis, strengthening quality assessment processes for Generative AI solutions and supporting production readiness.
- Operationalized machine learning lifecycle processes through MLflow, automated validation, CI/CD, model monitoring, and deployment controls, enabling reliable transitions from experimentation to production environments across enterprise workflows.
- Optimized scalable batch and streaming workloads with PySpark, Apache Spark, Kafka, and AWS, improving data-processing reliability for near real-time machine learning consumption across enterprise analytics workflows.
- Implemented Python-based REST and FastAPI services for ML inference and AI application integration, applying unit testing and CI/CD practices to strengthen reliable production AI workflows and deployments.

Machine Learning Engineer | Intel Dec 2019 – July 2024 | India

- Developed predictive machine learning models using Python, SQL, Scikit-learn, and XGBoost for semiconductor manufacturing and product analytics, improving forecast accuracy by 18% across operational use cases.
- Identified manufacturing defects, equipment abnormalities, process deviations, and silicon-test performance patterns through classification and anomaly-detection models built with Python and Scikit-learn across complex production datasets.
- Forecasted demand, capacity requirements, inventory levels, and supply-chain trends through time-series modeling, historical production analysis, and machine learning techniques supporting operational planning and resource optimization initiatives.
- Strengthened model reliability through feature engineering, evaluation, error analysis, and root-cause investigation across manufacturing-focused machine learning applications, improving predictive performance and identifying model limitations for operational deployment.
- Extracted contextual information from unstructured engineering and operational content using NLP, embeddings, and semantic retrieval workflows, enabling automated knowledge discovery and efficient analysis of complex technical datasets.
- Productionized machine learning workflows by integrating feature pipelines, model validation, inference processes, and analytical outputs into operational environments, coordinating with Data Engineering, MLOps, Product, and Supply Chain teams.

PROJECTS

Enterprise AI Knowledge Assistant | Python, LangGraph, LLMs, RAG, Pinecone, FastAPI

- Designed an enterprise knowledge assistant with Python, RAG, embeddings, and Pinecone, establishing document ingestion, pre-processing, indexing, and retrieval components for technical, operational, and business content.
- Orchestrated multi-step agent workflows with LangGraph and tool-based routing, enabling dynamic task decomposition, retrieval selection, source validation, and context-aware response generation for complex user requests.
- Optimized document chunking, metadata filtering, semantic retrieval, and prompt strategies, reducing irrelevant search results by 24% while improving contextual response consistency across evaluation scenarios.

LLM Evaluation & Guardrails Framework | Python, RAG, LLM Evaluation, Prompt Engineering, AWS

- Developed a Python-based LLM evaluation framework to measure retrieval relevance, response grounding, faithfulness, consistency, and overall generated answer quality across representative evaluation scenarios.
- Automated evaluation datasets and scoring workflows to identify hallucinations, retrieval failures, prompt weaknesses, and inconsistent outputs across multiple RAG application configurations and model responses.
- Implemented prompt validation and response guardrail checks to detect unsupported claims and policy-sensitive outputs, strengthening production quality controls and improving reliability across evaluated GenAI workflows.

EDUCATION

Master of Science in Data Science Aug 2024 - May 2026 | USA

Wichita State University, Kansas | Wichita, KS